

Interacting Multipoles of Rank 2 and 3

$$\begin{aligned}
 -\frac{1}{45} \cdot T_{\alpha\beta\gamma\delta\epsilon} \Theta_{\alpha\beta}^B \Omega_{\gamma\delta\epsilon}^A = R_z^{-11} \cdot -\frac{1}{45} \cdot & \left[\begin{aligned}
 & -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta(\delta, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta(\gamma, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta(\beta, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \delta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \gamma) \cdot \delta(\delta, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \delta) \cdot \delta(\gamma, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \epsilon) \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \gamma) \cdot \delta(\delta, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \delta) \cdot \delta(\gamma, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \epsilon) \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \beta) \cdot \delta(\delta, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \epsilon) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
 & -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A
 \end{aligned} \right] \quad (1)
 \end{aligned}$$

Simplify deltas:

$$\begin{aligned}
& -\frac{1}{45} \cdot T_{\alpha\beta\gamma\delta\epsilon} \Theta_{\alpha\beta}^B \Omega_{\gamma\delta\epsilon}^A = R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{aligned}
& -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta(\delta, \delta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\alpha}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\alpha}^B \cdot \Omega_{\beta\beta\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\alpha}^B \cdot \Omega_{\beta\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\alpha}^B \cdot \Omega_{\beta\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\epsilon}^A
\end{aligned} \right] \quad (2a)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{aligned}
& -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\delta\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\gamma\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\delta}^A \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\delta\epsilon}^A \\
& +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha}^B \cdot \Omega_{\gamma\delta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\alpha \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\beta\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\beta \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\gamma\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \Theta_{\alpha\alpha}^B \cdot \Omega_{\beta\beta\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\gamma \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\gamma}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \Theta_{\alpha\alpha}^B \cdot \Omega_{\beta\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\delta}^A \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \Theta_{\alpha\alpha}^B \cdot \Omega_{\beta\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\epsilon}^A \\
& -15 \cdot R_z^4 \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{\alpha\beta\epsilon}^A
\end{aligned} \right] \tag{2b}
\end{aligned}$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{array}{c} +315 \cdot R_z^5 \cdot \Theta_{z\beta}^B \cdot \Omega_{zz\beta}^A + 315 \cdot R_z^5 \cdot \Theta_{z\alpha}^B \cdot \Omega_{zz\alpha}^A \\ -45 \cdot R_z^5 \cdot \Theta_{z\beta}^B \cdot \Omega_{\beta\gamma\gamma}^A - 45 \cdot R_z^5 \cdot \Theta_{z\alpha}^B \cdot \Omega_{\alpha\gamma\gamma}^A \\ -90 \cdot R_z^5 \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{z\alpha\beta}^A \end{array} \right] \quad (3b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{array}{c} +315 \cdot R_z^5 \cdot \Theta_{z\alpha}^B \cdot \Omega_{zz\alpha}^A + 315 \cdot R_z^5 \cdot \Theta_{z\alpha}^B \cdot \Omega_{zz\alpha}^A \\ -45 \cdot R_z^5 \cdot \Theta_{z\beta}^B \cdot \Omega_{\alpha\alpha\beta}^A - 45 \cdot R_z^5 \cdot \Theta_{z\alpha}^B \cdot \Omega_{\alpha\beta\beta}^A \\ -90 \cdot R_z^5 \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{z\alpha\beta}^A \end{array} \right] \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{array}{c} +630 \cdot R_z^5 \cdot \Theta_{z\alpha}^B \cdot \Omega_{zz\alpha}^A - 45 \cdot R_z^5 \cdot \Theta_{z\beta}^B \cdot \Omega_{\alpha\alpha\beta}^A \\ -45 \cdot R_z^5 \cdot \Theta_{z\alpha}^B \cdot \Omega_{\alpha\beta\beta}^A - 90 \cdot R_z^5 \cdot \Theta_{\alpha\beta}^B \cdot \Omega_{z\alpha\beta}^A \end{array} \right] \quad (3d)$$

Simplify Quadrupoles:

$$-\frac{1}{45} \cdot T_{\alpha\beta\gamma\delta\epsilon} \Theta_{\alpha\beta}^B \Omega_{\gamma\delta\epsilon}^A = R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{array}{c} +630 \cdot R_z^5 \cdot \Theta_{zz}^B \cdot \Omega_{zzz}^A - 45 \cdot R_z^5 \cdot \Theta_{zz}^B \cdot \Omega_{z\alpha\alpha}^A \\ -45 \cdot R_z^5 \cdot \Theta_{zz}^B \cdot \Omega_{z\beta\beta}^A - 90 \cdot R_z^5 \cdot \Theta_{\alpha\alpha}^B \cdot \Omega_{z\alpha\alpha}^A \end{array} \right] \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{45} \cdot [+0] \quad (4b)$$

Combining everything:

$$-\frac{1}{45} \cdot T_{\alpha\beta\gamma\delta\epsilon} \Theta_{\alpha\beta}^B \Omega_{\gamma\delta\epsilon}^A = +0 \quad (5)$$